

■ A.2.3 Permutations

You can represent a permutation in *Mathematica* as a list of integers. The built-in function `Permutations` gives all possible permutations of lists of objects.

```
PermutationQ::usage = "PermutationQ[e] yields True if e represents a permutation."
```

```
PermutationQ[e_] := TrueQ[ Sort[e] == Range[Length[e]] ]
```

```
RandomPermutation::usage =  
  "RandomPermutation[n] gives a random permutation of n elements."
```

```
RandomPermutation[n_Integer?Positive] :=  
  Block[{t, i},  
    t = Table[{Random[], i}, {n}] ;  
    t = Sort[t];  
    Map[ Last, t ]  
  ]
```

Functions for checking permutations, and generating random ones.

```
Read in the permutations package.      In[1]:= <<Permutations.m  
                                       Out[1]= Permutations`
```

```
The number 2 appears twice, so this is not      In[2]:= PermutationQ[{3, 1, 2, 2}]  
considered a legal permutation.              Out[2]= False
```

```
Here is a random permutation of eight          In[3]:= RandomPermutation[8]  
elements.                                     Out[3]= {2, 3, 8, 1, 5, 4, 7, 6}
```

In many algorithms, it is necessary to write permutations as compositions of special cyclic permutations. Here are some functions that convert into and out of this form.

```
ToCycles::usage =  
  "ToCycles[p] writes the permutation p as a list of cyclic permutations."
```

```
ToCycles[perm_?PermutationQ] :=  
  Block[{a, t, n, l, i, len},  
    len = Length[perm];  
    a = {} ;  
    t = Table[True, {len}];  
    For[i=1, i<=len, i++,  
      If[t[[i]],  
        For[n = perm[[i]]; l = {},
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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```

t[[n]],
n = perm[[n]],
t[[n]] = False; AppendTo[l, n]
];
AppendTo[a, l]
]
];
Return[a]
]

```

```

FromCycles::usage =
  "FromCycles[{p1,p2,...}] gives the permutation that corresponds to
  a list of cycles."

```

```

FromCycles[cyc_List] :=
  Block[{list},
    Scan[ FromCycles0[list, #] &, cyc, 1] ;
    Array[list, Length[Flatten[cyc]]]
  ]

```

```

FromCycles0[list_, c_] :=
  Block[{c1, i},
    c1 = RotateRight[c];
    Table[ list[ c1[[i]] ] = c[[i]], {i, Length[c]} ]
  ]

```

Cycle forms of permutations.

Here is a random permutation of ten elements.

```

In[4]:= RandomPermutation[10]
Out[4]= {7, 1, 3, 8, 2, 9, 4, 10, 5, 6}

```

Here is the permutation written as a composition of cyclic permutations.

```

In[5]:= ToCycles[%]
Out[5]= {{7, 4, 8, 10, 6, 9, 5, 2, 1}, {3}}

```

This converts back from the cycle representation.

```

In[6]:= FromCycles[%]
Out[6]= {7, 1, 3, 8, 2, 9, 4, 10, 5, 6}

```